Summary in Graph

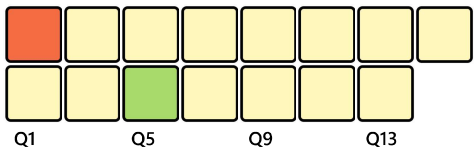
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 3)

Qs. Attempted:	**4** (4 + 0)	**Correct Marks:**	**1** (1 + 0)
Correct Attempts:	**1** (1 + 0)	**Penalty Marks:**	**0.33** (0.33 + 0)
Incorrect Attempts:	**3** (3 + 0)	**Resultant Marks:**	**0.67** (0.67 + 0)

Total Questions:	**15** (10 + 5)
Total Marks:	**20** (10 + 10)
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (43) out of 431 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

The soft margin SVM formulation is as follows:

$$\min \frac{1}{2}\mathbf{w}^T\mathbf{w} + C\sum_{i=1}^{N}\xi_i$$
$$\text{subject to } y_i\left(\mathbf{w}^T\mathbf{x_i} + b\right) \geq 1 - \xi_i \quad \forall i$$
$$\xi_i \geq 0 \quad \forall i$$

What is the behavior of the width of the margin $\left(\frac{2}{\|w\|}\right)$ as $C \to 0$?

A. Behaves like hard margin
B. Goes to zero
C. Goes to infinity
D. None of the above

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 02min 41sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

Which of the following are true of support vector machines?

A. Increasing the hyperparameter C tends to decrease the training error
B. The hard-margin SVM is a special case of the softmargin with the hyperparameter C set to zero
C. Increasing the hyperparameter C tends to decrease the margin
D. Increasing the hyperparameter C tends to decrease the sensitivity to outliers

Your Answer: A;B;D | Correct Answer: A;C | Incorrect | Time taken: 00min 50sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

You are training a soft-margin SVM on a binary classification problem. You find that your model's training accuracy is very high, while your validation accuracy is very low. Which of the following are likely to improve your model's performance on the validation data?

A. Training your model on more data.
B. Adding a quadratic feature to each sample point.
C. Increasing the hyperparameter C.
D. Decreasing the hyperparameter C.

Your Answer: A;B;C | Correct Answer: A;D | Incorrect | Time taken: 04min 02sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Suppose your training set for two-class classification in one dimension ($d = 1; x_i \in \mathbb{R}$) contains three sample points: point $x_1 = 3$ with label $y_1 = 1$, point $x_2 = 1$ with label $y_2 = 1$, and point $x_3 = -1$ with label $y_3 = -1$. What are the values of w and b given by a **hard-margin SVM**?

A. $w = 1, b = 1$
B. $w = 0, b = 1$
C. w = 1, b = 0
D. w = ∞, b = 0

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 34min 45sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

With a soft-margin SVM, which samples will have non-zero slack variables ξ_i ?

A. All misclassified samples
B. All samples lying on the margin boundary

C. All samples inside the margin
D. All samples outside the margin

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 02min 07sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Recall the soft-margin SVM objective function $|\mathbf{w}|^2 + C\sum_i \xi_i$. Which value of C is most likely to overfit the training data?

A. $C = 0.01$
B. $C = 0.00001$
C. $C = 1$
D. $C = 100$

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 33sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Which the following facts about the 'C' in SVMs is true?

A. As C approaches 0, the soft margin SVM is equal to the hard margin SVM.
B. A larger C-tends to create a larger margin.
C. C can be negative, as long as each of the slack variables are nonnegative
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 01sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

The minimum distance between two points of opposite classes in a hard margin Support Vector Machine is

A. $\frac{|w\|^2}{2}$
B. $\frac{1}{\|w\|^2}$
C. $\frac{2}{\|w\|}$
D. $\frac{2}{\|\boldsymbol{w}\|^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Let $\mathbf{w} = (1, 2)$ and $b = 3$. For the point $\mathbf{x} = (4, 5)$ with label $y = -1$, what is the smallest slack value ξ required for it to satisfy the margin constraint?

Your Answer: | Correct Answer: 18 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a small dataset with four points, where each point is in 2D:

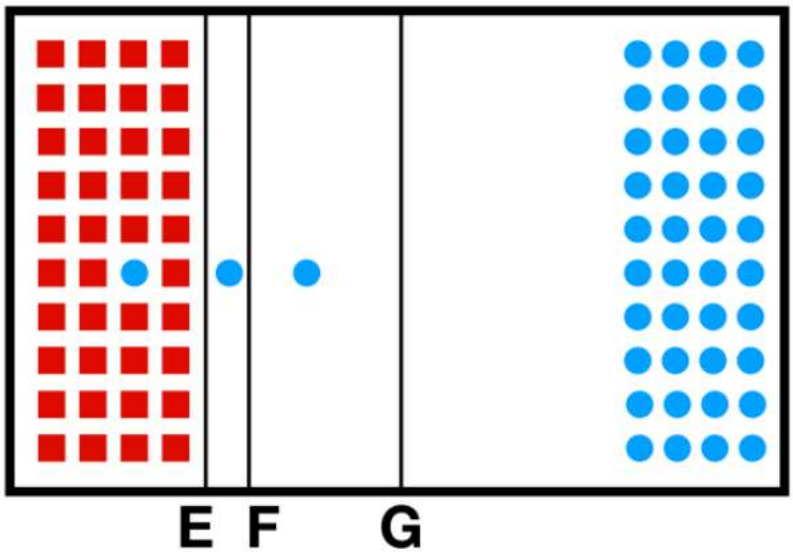
$$(x_{11} = 0, x_{12} = 0), y_1 = 0$$
$$(x_{21} = 0, x_{22} = 1), y_2 = 1$$
$$(x_{31} = 1, x_{32} = 0), y_3 = 1$$
$$(x_{41} = 1, x_{42} = 1), y_4 = 0$$

Which of the following statements is true regarding the classification of this dataset using a linear Support Vector Machine (SVM)? A) A linear SVM can perfectly classify this dataset. B) A linear SVM cannot perfectly classify this dataset because it is not linearly separable. C) Mapping each feature vector $\mathbf{X} = (x_1, x_2)$ into $\phi(\mathbf{X}) = (x_1 + x_2, x_1 - x_2)$ will make the dataset linearly separable. D) None of the above.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Soft SVM is run on the following training set three times, with three different settings of the hyperparameter C (the slack penalty weight). The first run with $C = C_E$ gives decision boundary E. The second, with $C = C_F$, gives F. The third, with $C = C_G$, gives G.
Which of the following correctly sorts C_E, C_F, and C_G by magnitude?



A. $C_E < C_F < C_G$
B. $C_G < C_F < C_E$
C. $C_E = C_F = C_G$
D. $C_F < C_G < C_E$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose that after solving a soft margin SVM problem, we obtain that the best separating hyperplane is $w^T x + b = 0$ for $w = [1, -2]$ and $b = 3$. Consider the following points $x_1 = [2, 1], x_2 = [-0.5, 1.5]$, and $x_3 = [-1.75, 0.5]$.
What are the labels (+1 or -1) assigned by our model to the three points?

A. $y_1 = +1, y_2 = +1, y_3 = -1$
B. $y_1 = -1, y_2 = +1, y_3 = +1$
C. $y_1 = +1, y_2 = -1, y_3 = +1$
D. $y_1 = -1, y_2 = -1, y_3 = +1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

$$\min_{\mathbf{w},\{\xi_i\}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N}\xi_i$$
$$\text{subject to } y_i \cdot \left(\mathbf{w}^T\mathbf{x}_i\right) \geq 1 - \xi_i, \forall i \text{ and } \xi_i \geq 0$$

where $\mathbf{w}$ are trainable parameters (including the bias term) and $\mathbf{x}_\mathrm{i}$ is the i -th data sample. Which of the following statements regarding this formulation are true?
1. If $0 < \xi_i \leq 1$, sample i lies inside the margin.
2. If $0 < \xi_i \leq 1$, sample i is correctly classified.
3. If $\xi_i \geq 1$, we cannot tell whether the sample i was correctly classified.
4. Setting C to 0 would prevent the SVM from misclassifying any sample.
5. The bigger the C, the less misclassifications will the SVM make.
6. The formulation above results in a strictly linear decision boundary in the original space.
Which of the statements are true?

A. $1, 2, 5, 6$
B. $1, 2, 3, 4, 6$
C. $1, 2, 3, 6$
D. $2, 4, 6$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

An SVM is trained with a linear kernel to do hard margin classification, i.e. the slack variables are not used, no misclassifications are allowed and the formulation is simply

$$\min_{\mathrm{wit}}, \frac{1}{2}\|\mathbf{w}\|^2$$
$$\text{subject to } y_i \cdot \left(\mathbf{w}^T\mathbf{x_i}\right) \geq 1, \forall i$$

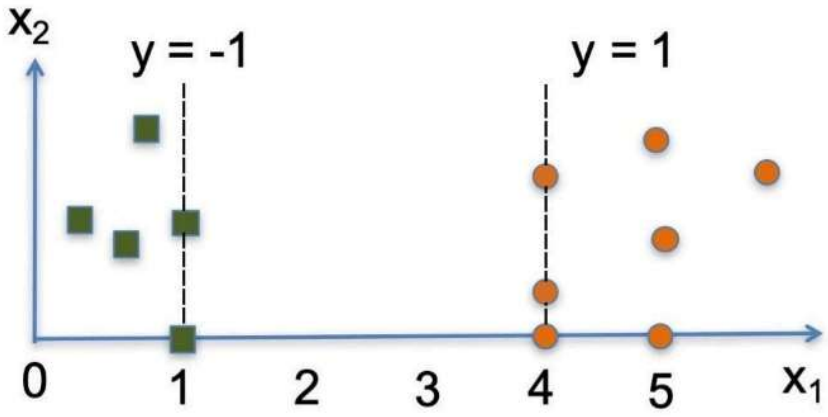
We know the support vectors and the margins after the training. Now, we introduce a new data point and retrain. Select the correct statements:

A. The margin will never change if the new data point is on the correct side of the original decision boundary and inside the margin.
B. The margin will not change if the new data point is on the correct side of the original decision boundary and outside of the margin.
C. The margin will change if the new data point is in between the original margin regardless of its positioning with respect to the decision boundary.
D. The margins will not change if the new data point is within the original margin and on the correct side of the original decision boundary.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

What will be the decision boundary learnt by solving the SVM optimization problem?



A. $\mathbf{w} = [2.5, 0]^\top$, and $b = 0$
B. $\mathbf{w} = [2/3, 0]^\top$, and $b = -5/3$.
C. $\mathbf{w} = [1, 0]^\top$, and $b = -2.5$.
D. $\mathbf{w} = [5/3, 0]^\top$, and $b = 2/3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff